YOSO: Complete Experimental Results

完整实验结果表

Overall localization

整体定位效果

All localization metrics are planar errors in metres. Smaller values indicate lower error.

定位指标均为平面误差，单位为米；数值越小，误差越低。

Table 1. Main comparison across all three groups.

Method	Median (m)	Mean (m)	P90 (m)	P95 (m)
I. Wi-Fi only, without calibration				
WKNN	2.329	2.547	4.472	5.179
RF	2.105	2.381	4.363	5.123
EWOk	1.938	2.194	3.981	4.712
DeepMetricFi	1.970	2.243	4.036	4.855
SSD-WKNN	2.141	2.351	4.117	4.782
II. Sensor fusion, without calibration				
WKNN + CS/IMU (PF)	1.407	1.590	2.794	3.530
RF + CS/IMU (PF)	1.300	1.429	2.541	2.893
EWOk + CS/IMU (PF)	1.200	1.344	2.356	2.767
DeepMetricFi + CS/IMU (PF)	1.272	1.381	2.316	2.663
SSD + CS/IMU (PF)	1.279	1.447	2.452	3.009
CS only (PF)	1.794	2.155	3.757	5.382
CS + IMU (PF)	1.450	1.848	3.588	4.482
III. Cross-device calibration / transfer				
Linear calibration + PF	1.396	1.557	2.798	3.290
ULFT	0.922	1.134	2.171	2.668
TCA	0.963	1.293	2.470	3.261
MetaLoc	0.995	1.383	2.806	3.571
WiDAGCN	1.045	1.266	2.467	3.056
iToLoc	0.885	1.102	2.122	2.684
YOSO	0.588	0.721	1.444	1.701

Component ablation and AP relocation

组件消融与 Wi-Fi AP 位置移动

Table 2. Main component ablation.

Method	Median (m)	Mean (m)	P90 (m)	P95 (m)	RSSI MAE (dB)
Full	**0.588**	**0.721**	**1.444**	**1.701**	3.614
A1: Joint-Region Scoring	0.860	1.280	2.870	3.519	3.788
A2: Self Fit	0.850	1.265	2.672	3.688	3.759
A3: w/o Wi-Fi-to-CS Feedback	0.645	0.800	1.623	1.966	3.627
A4: w/o Spatially Conditioned Device Response	0.734	0.844	1.532	2.034	**3.553**
A5: w/o Spatial Discriminative Readout	0.712	0.913	1.746	2.051	3.630

Table 3. All completed AP-relocation comparisons and RF configurations.

Method	Median (m)	Mean (m)	P90 (m)	P95 (m)
YOSO	**0.834**	**1.059**	**1.877**	**2.485**
EWOk	2.825	3.010	5.495	6.061
DeepMetricFi	2.605	2.853	5.232	5.911
ULFT	1.332	1.488	2.705	3.093
TCA	1.585	1.786	3.201	3.962
MetaLoc	1.957	2.181	3.857	4.711
WiDAGCN	1.681	1.934	3.516	4.463
iToLoc	1.517	1.698	3.085	3.782
RF (source)	2.751	2.907	5.148	5.793
RF (native)	2.707	2.857	5.034	5.615
RF-OPPO	1.609	1.767	3.146	3.758
RF-five	1.727	1.895	3.316	4.050
RF-covariance	1.556	1.650	2.713	3.111

CS anchor-count comparison

CS 锚点数量变化下的完整对比

For each anchor count, metrics are averaged equally across all same-size anchor subsets.
同一锚点数量下，各指标对全部锚点子集等权取平均。

Table 4. 1 CS anchor.

Method	Median (m)	Mean (m)	P90 (m)	P95 (m)
YOSO	**0.868**	**1.044**	**1.970**	**2.400**
A1: joint-region scoring	1.690	2.043	3.860	5.182
ULFT	1.243	1.435	2.734	3.245
TCA	1.528	1.841	3.570	4.481
MetaLoc	1.482	1.667	3.145	3.685
WiDAGCN	1.510	1.711	3.200	3.703
iToLoc	1.138	1.341	2.493	3.187
EWOk	1.938	2.194	3.981	4.712
DeepMetricFi	1.970	2.243	4.036	4.855
CS only (PF)	2.309	2.464	4.279	4.799
CS+IMU (PF)	1.995	2.299	4.444	5.130

Table 5. 2 CS anchors.

Method	Median (m)	Mean (m)	P90 (m)	P95 (m)
YOSO	**0.840**	**0.961**	**1.819**	**2.140**
A1: joint-region scoring	1.376	1.803	3.817	4.602
ULFT	1.134	1.346	2.563	3.153
TCA	1.258	1.557	3.083	3.931
MetaLoc	1.258	1.531	3.064	3.903
WiDAGCN	1.292	1.562	2.939	3.746
iToLoc	1.063	1.290	2.447	3.138
EWOk	1.938	2.194	3.981	4.712
DeepMetricFi	1.970	2.243	4.036	4.855
CS only (PF)	1.982	2.235	4.098	4.901
CS+IMU (PF)	1.734	2.126	4.253	5.204

CS anchor-count comparison

CS 锚点数量变化下的完整对比

For each anchor count, metrics are averaged equally across all same-size anchor subsets.
同一锚点数量下，各指标对全部锚点子集等权取平均。

Table 6. 3 CS anchors.

Method	Median (m)	Mean (m)	P90 (m)	P95 (m)
YOSO	**0.700**	**0.840**	**1.605**	**2.027**
A1: joint-region scoring	1.108	1.478	3.186	3.961
ULFT	0.997	1.227	2.423	2.988
TCA	1.077	1.345	2.584	3.382
MetaLoc	1.114	1.370	2.641	3.351
WiDAGCN	1.174	1.442	2.752	3.419
iToLoc	0.982	1.184	2.248	2.820
EWOk	1.938	2.194	3.981	4.712
DeepMetricFi	1.970	2.243	4.036	4.855
CS only (PF)	1.861	2.204	4.163	5.197
CS+IMU (PF)	1.583	2.012	4.053	4.960

Table 7. 4 CS anchors.

Method	Median (m)	Mean (m)	P90 (m)	P95 (m)
YOSO	**0.588**	**0.721**	**1.444**	**1.701**
A1: joint-region scoring	0.860	1.280	2.870	3.519
ULFT	0.922	1.134	2.171	2.668
TCA	0.963	1.293	2.470	3.261
MetaLoc	0.995	1.383	2.806	3.571
WiDAGCN	1.045	1.266	2.467	3.056
iToLoc	0.885	1.102	2.122	2.684
EWOk	1.938	2.194	3.981	4.712
DeepMetricFi	1.970	2.243	4.036	4.855
CS only (PF)	1.794	2.155	3.757	5.382
CS+IMU (PF)	1.450	1.848	3.588	4.482

Wi-Fi update rate

Wi-Fi 更新频率变化

Table 8. Wi-Fi cap: 0.25 Hz.

Method	Median (m)	Mean (m)	P90 (m)	P95 (m)
YOSO	**0.808**	**0.937**	**1.687**	**2.099**
Linear calibration + PF	1.462	1.837	3.486	4.129
ULFT	1.264	1.526	2.742	3.792
TCA	1.025	1.267	2.469	3.144
MetaLoc	1.159	1.537	2.981	3.929
WiDAGCN	1.277	1.693	3.134	4.548
iToLoc	1.178	1.432	2.657	3.406

Table 9. Wi-Fi cap: 0.5 Hz.

Method	Median (m)	Mean (m)	P90 (m)	P95 (m)
YOSO	**0.691**	**0.914**	**1.826**	**2.379**
Linear calibration + PF	1.437	1.755	3.371	3.915
ULFT	1.191	1.451	2.738	3.390
TCA	1.014	1.259	2.487	3.420
MetaLoc	1.215	1.582	3.156	4.183
WiDAGCN	1.178	1.539	2.946	4.180
iToLoc	1.152	1.433	2.771	3.490

Wi-Fi update rate

Wi-Fi 更新频率变化

Table 10. Wi-Fi cap: 1 Hz.

Method	Median (m)	Mean (m)	P90 (m)	P95 (m)
YOSO	**0.696**	**0.911**	**1.744**	**2.233**
Linear calibration + PF	1.430	1.799	3.426	4.147
ULFT	1.067	1.331	2.554	3.385
TCA	0.983	1.226	2.409	3.146
MetaLoc	1.122	1.634	3.391	5.032
WiDAGCN	1.146	1.555	3.089	4.258
iToLoc	1.111	1.334	2.600	3.153

Table 11. Wi-Fi cap: 2 Hz.

Method	Median (m)	Mean (m)	P90 (m)	P95 (m)
YOSO	**0.713**	**0.863**	**1.664**	**2.072**
Linear calibration + PF	1.414	1.664	3.052	3.546
ULFT	1.026	1.249	2.372	2.968
TCA	0.942	1.161	2.396	2.895
MetaLoc	1.015	1.341	2.579	3.591
WiDAGCN	1.108	1.454	2.885	3.689
iToLoc	0.953	1.175	2.193	2.903

Cross-site transfer, initialization, and on-device updates

跨站点迁移、初始化与逐次端侧更新

Table 12. Cross-site transfer with zero parameter updates.

Method	Median (m)	Mean (m)	P90 (m)	P95 (m)
Map1 (original)	0.588	0.721	1.444	1.701
Map2 (zero updates)	0.760	0.786	1.388	1.429
Map3 (zero updates)	0.649	0.815	1.472	1.750

Table 13. Start error and subsequent localization.

Start setting	Initial median (m)	Tracking median (m)	Tracking mean (m)	Tracking P90 (m)	Tracking P95 (m)
Known start	0.000	0.588	0.721	1.444	1.701
5 s stationary	0.785	0.697	0.789	1.419	1.648
2 s stationary	1.051	0.717	0.862	1.676	2.016

Table 14. Per-update end-to-end computation and resident memory.

Method	Median latency (ms)	P95 latency (ms)	Peak RSS (MB)
ULFT	7.59	8.89	663.10
TCA	20.01	24.56	829.10
MetaLoc	15.06	23.27	461.40
WiDAGCN	1375.09	1526.03	940.40
iToLoc	954.17	981.93	452.10
YOSO	561.80	844.09	479.20

Representative-route comparisons

代表性路线的定量对比

These statistics describe the fixed route shown by the trajectory panels, using the original prediction points.

本组统计对应轨迹图中的固定路线，使用原始预测点计算。

Table 15. Representative route with 1 CS anchor.

Method	Median (m)	Mean (m)	P90 (m)	P95 (m)
YOSO	**0.655**	**0.702**	**1.181**	**1.252**
EWOk + CS/IMU (PF)	1.159	1.192	2.223	2.434
Linear calibration + PF	1.217	1.225	2.008	2.202
ULFT	1.341	1.320	2.039	2.298
TCA	1.437	1.630	2.935	3.249
MetaLoc	1.315	1.361	2.031	2.300
WiDAGCN	1.172	1.278	2.161	2.524
iToLoc	1.017	1.043	1.673	1.806

Table 16. Representative route with 2 CS anchors.

Method	Median (m)	Mean (m)	P90 (m)	P95 (m)
YOSO	**0.557**	**0.650**	**1.164**	1.613
EWOk + CS/IMU (PF)	1.223	1.255	2.157	2.358
Linear calibration + PF	1.237	1.252	2.051	2.278
ULFT	0.882	0.947	1.435	**1.588**
TCA	0.912	0.958	1.551	1.697
MetaLoc	0.937	1.101	2.214	2.619
WiDAGCN	0.940	1.028	1.863	2.150
iToLoc	1.026	1.091	1.800	2.095

Representative-route comparisons

代表性路线的定量对比

These statistics describe the fixed route shown by the trajectory panels, using the original prediction points.

本组统计对应轨迹图中的固定路线，使用原始预测点计算。

Table 17. Representative route with 3 CS anchors.

Method	Median (m)	Mean (m)	P90 (m)	P95 (m)
YOSO	**0.474**	**0.521**	**0.905**	**1.005**
EWOk + CS/IMU (PF)	1.116	1.168	2.023	2.253
Linear calibration + PF	1.147	1.230	2.234	2.369
ULFT	0.936	0.895	1.507	1.608
TCA	0.922	0.978	1.730	1.834
MetaLoc	0.988	1.135	2.231	2.707
WiDAGCN	0.929	1.006	1.763	2.174
iToLoc	0.931	1.032	1.745	2.031

Table 18. Representative route with 4 CS anchors.

Method	Median (m)	Mean (m)	P90 (m)	P95 (m)
YOSO	**0.451**	**0.517**	**0.899**	**1.089**
EWOk + CS/IMU (PF)	0.958	1.073	1.945	2.313
Linear calibration + PF	1.104	1.272	2.429	2.956
ULFT	0.862	0.977	1.723	1.937
TCA	0.856	0.911	1.662	1.826
MetaLoc	0.923	1.092	2.072	2.610
WiDAGCN	0.792	0.924	1.745	2.073
iToLoc	0.824	0.865	1.449	1.773

Sparse-anchor trained component variants

稀疏锚点下的独立训练组件变体

The joint-fit and self-fit rows here are independently trained variants. The main scoring interventions are reported in Table 2.

这里的联合拟合与自身拟合行对应独立训练的变体；主实验中的评分干预见表 2。

Table 19. Trained variants with 1 CS anchor.

Method	Median (m)	Mean (m)	P90 (m)	P95 (m)	RSSI MAE (dB)
YOSO	**0.868**	1.044	1.970	2.400	3.852
Joint-fit trained variant	0.892	1.051	1.992	2.388	3.813
Self-fit trained variant	0.890	**1.021**	**1.906**	**2.251**	3.814
A3	0.908	1.063	2.025	2.453	3.850
A4	0.905	1.037	1.934	2.332	**3.804**
A5	1.153	1.292	2.314	2.816	3.866

Table 20. Trained variants with 2 CS anchors.

Method	Median (m)	Mean (m)	P90 (m)	P95 (m)	RSSI MAE (dB)
YOSO	0.840	0.961	1.819	2.140	3.737
Joint-fit trained variant	0.842	0.992	1.880	2.231	3.806
Self-fit trained variant	**0.819**	0.970	1.836	2.205	3.807
A3	0.843	**0.953**	**1.777**	**2.128**	3.747
A4	0.825	0.958	1.806	2.207	**3.679**
A5	0.980	1.086	1.908	2.236	3.751

Table 21. Trained variants with 3 CS anchors.

Method	Median (m)	Mean (m)	P90 (m)	P95 (m)	RSSI MAE (dB)
YOSO	**0.700**	**0.840**	**1.605**	2.027	3.624
Joint-fit trained variant	0.735	0.895	1.672	2.071	3.801
Self-fit trained variant	0.729	0.899	1.746	2.119	3.804
A3	0.718	0.851	1.662	**1.933**	3.632
A4	0.755	0.909	1.731	2.198	**3.581**
A5	0.843	0.970	1.818	2.167	3.640

Trajectory-level replay runtime

整条轨迹回放的计算耗时

Runtime is measured per trajectory. Amortized time divides trajectory computation time by its output count. This is a trajectory-level measurement protocol.

耗时按整条轨迹测量；每点摊销耗时为轨迹计算时间除以输出数。本页采用轨迹级测量协议。

Table 22. Completed trajectory replay measurements.

Method	Trajectory median (ms)	Trajectory P95 (ms)	Amortized median (ms/point)	Amortized P95 (ms/point)
WKNN	1.623	2.243	0.027	0.030
RF	20.528	24.813	0.344	0.473
EWOk	157.233	222.468	2.530	2.738
DeepMetricFi	56.883	79.113	0.922	0.978
SSD-WKNN	3.474	4.751	0.057	0.061
WKNN + CS/IMU (PF)	88.926	114.453	1.392	1.557
RF + CS/IMU (PF)	104.898	129.693	1.683	1.850
EWOk + CS/IMU (PF)	240.237	332.175	3.873	4.203
DeepMetricFi + CS/IMU (PF)	142.007	187.655	2.253	2.479
SSD + CS/IMU (PF)	86.472	113.362	1.367	1.539
CS only (PF)	47.786	64.399	0.765	0.900
CS + IMU (PF)	73.875	97.329	1.162	1.286
Linear calibration + PF	1359.431	1849.699	21.845	24.854
ULFT	2144.520	3178.796	34.025	39.841
TCA	12653.446	15762.857	214.150	271.198
MetaLoc	1709.403	2230.152	26.438	34.816

Trajectory-replay memory and deployment assets

轨迹回放的内存与部署资源

PSS is proportional set size. Load time is measured separately from replay computation; deployment size covers method-specific assets.
PSS 表示按共享比例分摊的物理内存。加载时间与回放计算分别测量，部署大小对应方法所需的资源文件。

Table 23. Resources for the trajectory-replay measurements.

Method	Peak PSS (MB)	Deployment (MB)	Load time (ms)
WKNN	75.501	0.006	14.340
RF	151.730	3.755	1022.086
EWOk	76.903	0.075	27.180
DeepMetricFi	197.551	0.715	55.306
SSD-WKNN	123.842	4.825	424.825
WKNN + CS/IMU (PF)	98.082	0.006	347.787
RF + CS/IMU (PF)	153.426	3.755	804.777
EWOk + CS/IMU (PF)	99.801	0.075	349.250
DeepMetricFi + CS/IMU (PF)	203.349	0.715	170.706
SSD + CS/IMU (PF)	123.586	4.825	431.404
CS only (PF)	93.517	0.001	331.159
CS + IMU (PF)	93.788	0.001	340.185
Linear calibration + PF	249.852	7.250	197.744
ULFT	396.072	7.273	210.439
TCA	513.994	7.250	211.474
MetaLoc	305.923	7.672	235.971